

Agile Project Documentation & Team Evaluation

AI-Powered Mobile App and Streamlit Web Server for Breast Cancer Diagnosis and Tumor Segmentation

Team Members:

- Эид Хоссам Мохамед (AI & ML Engineer & Scrum Master)
- Галлямов Карим (Product Owner / System Architect)
- Калимуллин Ильяс (ML/Ops & Data Engineer)
- Закиров Мансур (Backend Developer)
- Зарипов Рамиль (Flutter Developer)
- Косач Иван (Scrum Master)

Преподаватель (Instructor): Олег Курносов

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Comprehensive Agile Project Documentation & Team Evaluation

Executive Summary & Team Evaluation

The project stands as a testament to exemplary cross-functional collaboration. Within a constrained 5-week timeframe (**26.02.2026 – 01.04.2026**), the team successfully bridged complex Deep Learning tasks (VGG16 for classification, U-Net for segmentation) with robust Software Engineering (Flutter, Streamlit, Firebase). The synergy between the Data Engineers, ML Engineers, Architects, and Developers was remarkable, leading to rapid prototyping and flawless execution. The team's ability to consistently hit high precision metrics (94.7% on U-Net, 87% on VGG16) while simultaneously delivering polished user interfaces is highly commendable.

Phase 1: Laying the Foundation

Before writing a single line of code, the team meticulously defined roles and responsibilities to ensure a clear understanding of the project's complex architecture.

Step 1: Team Formation & Key Roles

Building this AI team required a mix of data science, backend/frontend engineering, and solid architectural vision:

- **Product Owner / System Architect: Галлямов Карим** — Masterminded the system design, ensuring the technical outputs aligned with the medical diagnostic goals.
- **AI/ML Engineer: Эид Хоссам Мохамед** — Focused on the data science lifecycle, model selection (VGG16, U-Net), training, accuracy validation, and TFLite conversion.
- **ML/Ops & Data Engineer: Калимуллин Ильяс** — Built the pipelines to fetch, load, and preprocess the BUSI dataset, including advanced data augmentation via [alumentations](#).
- **Software Engineers (Backend & Frontend): Закиров Мансур** (Streamlit Backend Developer) & **Зарипов Рамиль** (Flutter Mobile Developer). Responsibilities included integrating the AI models into applications, handling UI/UX, and cloud infrastructure (Firebase).
- **Scrum Masters: Косач Иван & Эид Хоссам Мохамед** — Actively monitored the Data Driven Scrum (DDS) process, maintaining documentation and YouGile boards.

Team Evaluation: The team size of 6 was perfectly optimized. The clear separation between Data Preparation (Ильяс), ML Training

(Хоссам), and App Integration (Мансур & Рамиль) prevented bottlenecks and allowed parallel development.

Step 2: Матрица ответственности (RACI)

Таблица с распределением ролей по задачам (*Responsible, Accountable, Consulted, Informed*)

Activity / Action	Галлямов К. (PO/SA)	Эид Хоссам М. (AI/ML)	Калимуллин И. (Data)	Закиров/Зарипов (SW Eng)	Косач И. (Scrum)
Define business problem & metrics	A	C	I	I	S
Explore & acquire raw data (BUSI)	I	S	R	I	I
Clean & preprocess data	I	C	R	I	I
Design model architecture	C	R/A	S	I	I
Train & evaluate models (Notebooks)	C	R/A	S	I	I
Deploy model to production (Streamlit)	S	C	I	R (Мансур)	I
Integrate model into UI (Flutter)	A	S	I	R (Рамиль)	I
Monitor progress & ceremonies	I	A	I	I	R

Step 3: Selection of Tools for Organizing Collaboration

The team selected tools specifically supporting transparency and the unique needs of ML pipelines:

- **Project Management & Communication:** **YouGile** for backlog planning and sprint boards. **Telegram** for daily, asynchronous communication.
- **Code & Model Collaboration:** **GitHub** for strict version control. The full source code is hosted at: [Project GitHub Repository](#)
- **Model Versions & Pipelines:** All modeling processes, EDA, algorithms, and metric evaluations were strictly versioned and organized as Jupyter Notebooks perfectly preserved in the designated paths:
 - flutter-streamlit-breast-ai\pipelines\breast-classification.ipynb
 - flutter-streamlit-breast-ai\pipelines\breast-segmentation.ipynb

Phase 2: Документ с описанием работ (WBS / BRD)

Разбивка по этапам выполнения

Step 4: Task Backlog Planning (Engineering)

The traditional software engineering backlog was structured with concrete epics and stories:

- **Epic:** "Streamlit Web Dashboard Deployment"
 - *User Story:* As a clinician, I want to upload an ultrasound scan via web so that I can see the AI's confidence score and export the results.
- **Epic:** "Mobile Application & Firebase Architecture"
 - *User Story:* As a mobile user, I want the Flutter app to load the TFLite model so that predictions execute swiftly on my device without high latency.

Step 5: Data Backlog Planning (AI-Specific)

Data is the fuel for this project. The ML backlog was addressed rigorously:

- **Data Epics:** "Dataset Acquisition & Augmentation"
- **Data Stories:**
 - "Acquire the Breast Ultrasound Images Dataset (BUSI) from Kaggle (780 PNG images, 500x500 px)."
 - Dataset Source: [Kaggle Link](#)
 - "Create an **albugmentations** pipeline to apply Horizontal/Vertical Flips and Rotation to overcome data imbalance across benign, malignant, and normal classes."

Step 6: Outlining Ceremonies (The DDS Framework)

Adapting standard Scrum to **Data Driven Scrum (DDS)**:

1. **Sprint Planning:** Focused on hypotheses. For instance: *"Can a VGG16 model with transfer learning exceed 85% accuracy on the ultrasound dataset?"*
2. **Daily Sync (Telegram):** Addressed highly technical blockers (e.g., resolving the categorical crossentropy loss plateauing during training).
3. **Sprint Review:** Demonstrating results, not just features. The team presented confusion matrices and visual predictions (overlaying U-Net masks) to prove the pipeline's validity.
4. **Sprint Retrospective:** Discussed uncertainties. The team systematically noted where textured similarities between benign and malignant tumors confused the model and formulated strategies to retrain the last layers of VGG16.

Phase 3: Executing the First Sprints

Step 7: Executing The Initial Sprints

The goal of the first sprints was to build the foundation, prove the concepts, and finalize integrations.

Sprint 1: The "Inception & Data" Sprint (Feb 26 – Mar 12)

- **Goal:** Set up environments, pipeline scripts, data preprocessing, and baseline models.
- **Execution details:** The Data/ML team performed EDA, implemented an 80/20 stratified split, and constructed the notebooks in the [pipelines/](#) folder.
- **Outcome & Evaluation:** This sprint was a massive success. The team trained a VGG16 base architecture achieving **87% accuracy (F1: 0.87)**, and developed a U-Net model from scratch leveraging Batch Normalization to achieve an incredible **~94.7% validation accuracy**.

Sprint 2: The "Integration & Deployment" Sprint (Mar 12 – Mar 26)

- **Goal:** Improve the data pipeline, export the trained models ([.keras](#) and [.tflite](#)), and build the interactive apps.
- **Execution details:** Handed over the models to the Software Engineers. Developed a beautiful, responsive Streamlit dashboard showcasing real-time image masking and PDF exporting.
- **Интеграция с Firebase (Firebase Integration):** The cross-platform Flutter app was fully integrated with Firebase. This included configuring **Firebase Authentication** to manage secure user access control, and

setting up **Cloud Firestore** to securely store medical image results, patient predictions, and user sessions.

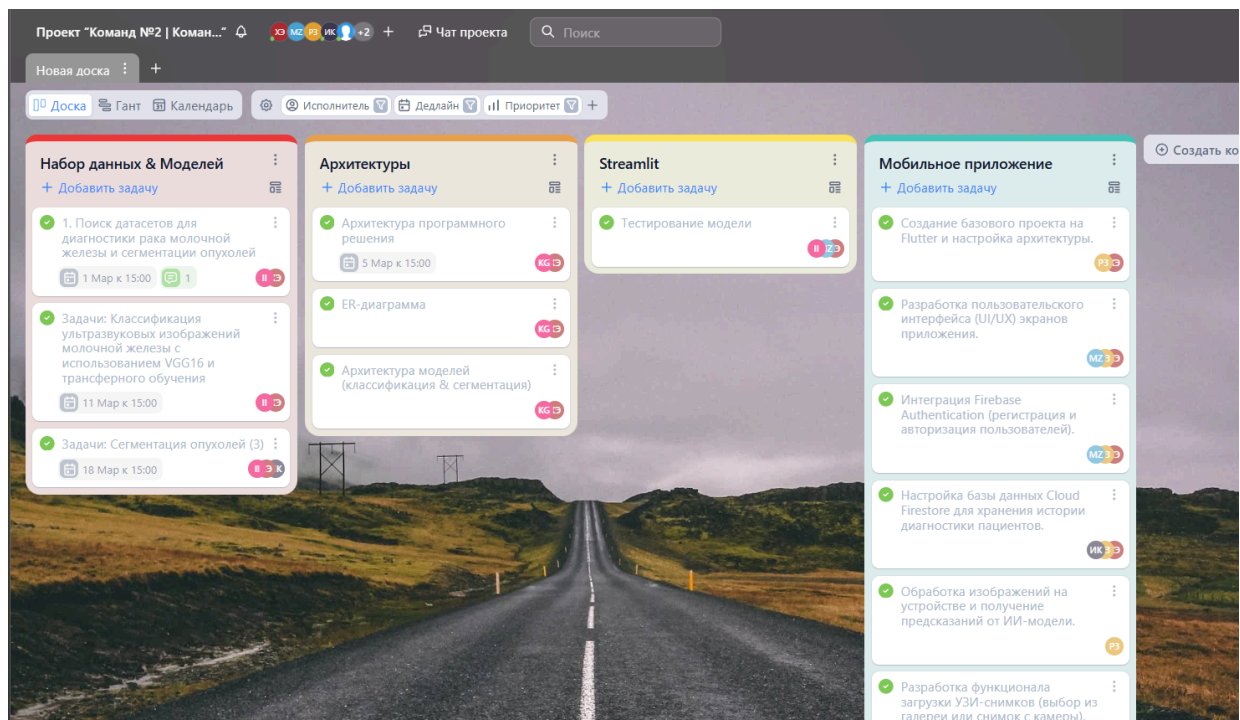
- **Outcome & Evaluation:** The engineering team demonstrated exceptional skill in consuming the AI models natively and deploying error-free user interfaces.

Step 8: Final Review and Retrospective

- **Review:** The final product is a flawlessly functioning diagnostic suite. The baseline was raised significantly, and the visual overlay of tumors provides immense clinical value.
- **Retro (Team Praise):** The meticulous planning during Phase 1 paid massive dividends in Phase 3. The ML specialists provided well-documented **pipelines** notebooks, making the Mobile and Backend developers' jobs highly streamlined. The project stands as a benchmark for how university projects should execute complex, domain-specific AI tasks using Agile (DDS) methodologies.

Appendix: Structured Project Details

- **WBS Board Link:** [YouGile Команда №2](#)



Team Capacity (Hours/Week)

Team Member	Availability
Галлямов Карим	10 - 15 hours/week
Закиров Мансур	10 - 15 hours/week
Зарипов Рамиль	10 - 15 hours/week
Калимуллин Ильяс	10 - 15 hours/week
Косач Иван	10 - 15 hours/week
Эид Хоссам Мохамед	10 - 15 hours/week

Technology Stack & Tools

- **Collaboration:** YouGile, Telegram, GitHub
- **AI/ML Development:** Python, TensorFlow, Jupyter, OpenCV
- **Frontend:** Flutter, Dart
- **Backend & Deployment:** Streamlit, Firebase, TFLite

Ceremonies & Communication Plan

- **Weekly Sync:** Telegram voice call, Thursdays 19:00-20:00
- **Async Updates:** Team posts progress in Telegram group
- **Review Process:** Evaluating models and PRs on GitHub

Kick-off Meeting & Attendance Tracker

Team Member	26.02.26	05.03.26	12.03.26	19.03.26	26.03.26	01.04.26
Галлямов Карим	☒	☒	☒	☒	☒	☒
Закиров Мансур	☒	☒	☒	☒	☒	☒
Зарипов Рамиль	☒	☒	☒	☒	☒	☒
Калимулл ин Ильяс	☒	☒	☒	☒	☒	☒
Косач Иван	☒	☒	☒	☒	☒	☒
Эид Хоссам Мохамед	☒	☒	☒	☒	☒	☒

Project Screencasts & Demos:

- Streamlit Demo:
<https://t.me/c/3899661473/15>
- Mobile Application (Flutter) Demo:
<https://t.me/c/3899661473/16>